APPENDIX A: DERIVATION OF MULTI-BS LEAST-SQUARES LOCALIZATION

After obtaining the angle information of potential Eves through the VA-MUSIC method at a single BS, in order to obtain the precise physical coordinates of the target in three-dimensional space, we introduce B collaborative BSs for joint localization. The known position coordinates of the b-th BS in the three-dimensional Cartesian coordinate system is $\mathbf{w}_b = [x_b, y_b, 0]^T$, and the true three-dimensional coordinates of the e-th potential Eve are unknown $\mathbf{w}_e[n] = [x_e[n], y_e[n], z_e[n]]^T$. According to the estimation results of the VA-MUSIC method earlier, the zenith and azimuth angles of the Eve e estimated by the b-th BS are denoted as $(\hat{\theta}_{b,e}[n], \hat{\varphi}_{b,e}[n])$. Based on the conversion relationship from spherical coordinates to Cartesian coordinates, the unit direction vector $\boldsymbol{\chi}_{b,e}[n]$ pointing from the b-th BS to Eve e can be expressed as:

$$\boldsymbol{\chi}_{b,e}[n] = \begin{bmatrix} \sin\hat{\theta}_{b,e}[n]\cos\hat{\varphi}_{b,e}[n] \\ \sin\hat{\theta}_{b,e}[n]\sin\hat{\varphi}_{b,e}[n] \\ \cos\hat{\theta}_{b,e}[n] \end{bmatrix} \tag{67}$$

where the direction vector satisfies the unit length, that is, $\|\boldsymbol{\chi}_{b,e}[n]\|_2^2 = 1$.

In an ideal noise free environment, all rays originating from the BS position $\mathbf{w}_b$ and extending along the direction vector $\boldsymbol{\chi}_{b,e}[n]$ should precisely intersect at the true position $\mathbf{w}_e[n]$ of the Eve. However, due to measurement and estimation errors, these rays are usually non planar straight lines in three-dimensional space and cannot intersect perfectly. Let $\mathbf{w} \in \mathbb{R}^3$ denote the optimization variable representing the Eve position. Therefore, we take the square of the orthogonal distance from the target position $\mathbf{w}$ to the line of sight ray of the b-th BS as the error measure. Define the orthogonal projection matrix $\mathbf{P}_{b,e}[n] \in \mathbb{R}^{3\times3}$ perpendicular to the direction vector $\boldsymbol{\chi}_{b,e}[n]$ as:

$$\mathbf{P}_{b,e}[n] = \mathbf{I}_3 - \boldsymbol{\chi}_{b,e}[n]\boldsymbol{\chi}_{b,e}^T[n] \tag{68}$$

where $\mathbf{I}_3$ is the identity matrix of 3×3. Due to the fact that the projection matrix is idempotent (i.e., $\mathbf{P}_{b,e}^T[n]\mathbf{P}_{b,e}[n] = \mathbf{P}_{b,e}[n]$), the squared Euclidean distance from the unknown target point $\mathbf{w}$ to the observation ray at the b-th BS can be written as:

$$\epsilon_b^2 = \|\mathbf{P}_{b,e}[n](\mathbf{w} - \mathbf{w}_b)\|_2^2 = (\mathbf{w} - \mathbf{w}_b)^T\mathbf{P}_{b,e}[n](\mathbf{w} - \mathbf{w}_b) \tag{69}$$

In order to estimate the optimal three-dimensional coordinates of Eve e, we construct a Least Squares (LS) optimization problem with the objective of minimizing the sum of squared distances from the target point to all B BS observation rays. Objective function $J_e(\mathbf{w}; n)$ is:

$$J_e(\mathbf{w}; n) = \sum_{b=1}^{B}(\mathbf{w} - \mathbf{w}_b)^T\mathbf{P}_{b,e}[n](\mathbf{w} - \mathbf{w}_b) \tag{70}$$

The corresponding optimization problem can be expressed as:

$$\hat{\mathbf{w}}_e^{\mathrm{LS}}[n] = \arg\min_{\mathbf{w}} J_e(\mathbf{w}; n) \tag{71}$$

To solve the above optimization problem, we take partial derivatives of the objective function $J_e(\mathbf{w}; n)$ with respect to the target position vector $\mathbf{w}$. Using the principles of matrix calculus, the partial derivative is:

$$\frac{\partial J_e(\mathbf{w}; n)}{\partial \mathbf{w}} = 2 \sum_{b=1}^{B} \mathbf{P}_{b,e}[n](\mathbf{w} - \mathbf{w}_b) \tag{72}$$

Make the partial derivative equal to the zero vector to find the global optimal solution:

$$\sum_{b=1}^{B} \mathbf{P}_{b,e}[n]\mathbf{w} - \sum_{b=1}^{B} \mathbf{P}_{b,e}[n]\mathbf{w}_b = \mathbf{0} \tag{73}$$

extracting the term containing the unknown variable $\mathbf{w}$ from the above equation yields a system of linear equations:

$$\left(\sum_{b=1}^{B} \mathbf{P}_{b,e}[n] \right) \mathbf{w} = \sum_{b=1}^{B} \mathbf{P}_{b,e}[n]\mathbf{w}_b \tag{74}$$

To simplify the expression, let the system matrix $\mathbf{A}_e[n] \in \mathbb{R}^{3\times3}$ and aggregate vector $\mathbf{c}_e[n] \in \mathbb{R}^{3\times1}$ be:

$$\mathbf{A}_e[n] = \sum_{b=1}^{B} \left(\mathbf{I}_3 - \boldsymbol{\chi}_{b,e}[n]\boldsymbol{\chi}_{b,e}^T[n] \right) \tag{75}$$

$$\mathbf{c}_e[n] = \sum_{b=1}^{B} \left(\mathbf{I}_3 - \boldsymbol{\chi}_{b,e}[n]\boldsymbol{\chi}_{b,e}^T[n] \right) \mathbf{w}_b \tag{76}$$

Finally, the estimated value of the least squares closed form solution for the three-dimensional position of Eve e is:

$$\hat{\mathbf{w}}_e^{\mathrm{LS}}[n] = \mathbf{A}_e^{-1}[n]\mathbf{c}_e[n] = \left(\sum_{b=1}^{B} \left(\mathbf{I}_3 - \boldsymbol{\chi}_{b,e}[n]\boldsymbol{\chi}_{b,e}^T[n] \right) \right)^{-1} \times \left(\sum_{b=1}^{B} \left(\mathbf{I}_3 - \boldsymbol{\chi}_{b,e}[n]\boldsymbol{\chi}_{b,e}^T[n] \right) \mathbf{w}_b \right) \tag{77}$$